Privacy-Preserving Federated Learning for Multi-Institutional Healthcare Data

Abstract

We present a federated learning framework tailored to multi-institutional clinical data, combining differential privacy with secure aggregation to enable collaborative model training without sharing raw patient records. We evaluate the framework on a diabetic retinopathy detection task across five simulated hospital sites.

1. Introduction

Healthcare data is highly sensitive and fragmented across institutions, making centralized model training legally and ethically difficult. Federated learning offers a promising alternative by training a shared global model across decentralized data silos while keeping raw data local. This paper addresses the specific privacy and heterogeneity challenges of clinical imaging data.

2. Methodology

We implement FedAvg with per-client differential privacy noise injection and secure aggregation via additive masking. Five simulated hospital sites each hold a non-IID partition of a diabetic retinopathy fundus image dataset (35,000 images total). Local models are ResNet-50 backbones fine-tuned for 5 local epochs per federated round, with 100 total communication rounds.

3. Results

The federated model achieved 89.3 percent AUC on held-out test data, within 2.1 points of a centrally-trained baseline (91.4 percent AUC), while providing (epsilon=3.2)-differential privacy guarantees per client. Communication overhead was reduced 40 percent via gradient compression.

4. Limitations

The simulated non-IID partitioning may not fully reflect real-world institutional heterogeneity, and our privacy budget analysis assumes an honest-but-curious server threat model, which may not hold against fully malicious aggregators.

5. Future Work

We plan to evaluate the framework on real multi-hospital consortiums, incorporate personalization layers for site-specific adaptation, and explore robustness to Byzantine clients.

References

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